Wednesday

1. Test 3 monday
   A. Study Guide on CourseDen
2. Finish Polar Area
   A. worksheet
3. Start 11.1

Friday

Review for test

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2. $r = 3e^{1.4\theta}$ Find horizontal tangent

Given $r = f(\theta)$, we treat θ as a parameter:

$$x = r\cos\theta \qquad \text{and} \qquad y = r\sin\theta$$

Replacing r with $f(\theta)$ in these gives us

$$x = f(\theta)\cos\theta \qquad \text{and} \qquad y = f(\theta)\sin\theta$$

From here we get

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$x = 3e^{1.4\theta}\cos\theta$ $\qquad$ $y = \underbrace{3e^{1.4\theta}}\underbrace{\sin\theta}$

$$\frac{dy}{d\theta} = 3e^{1.4\theta}\cos\theta + \sin\theta \cdot 1.4 \cdot 3e^{1.4\theta}$$

$$= 3e^{1.4\theta}\left[\cos\theta + 1.4\sin\theta\right] = 0$$

$3e^{1.4\theta} = 0$

No solution

$\cos\theta + 1.4\sin\theta = 0$

$\cos\theta = -1.4\sin\theta$

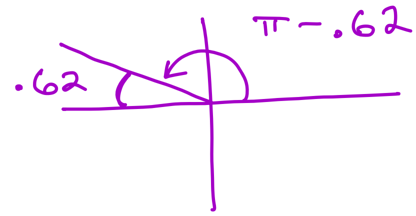
$1 = -1.4\dfrac{\sin\theta}{\cos\theta}$

$$\frac{1}{-1.4} = \frac{-1.4\tan\theta}{-1.4}$$

$$-\frac{5}{7} = \tan\theta$$

$$\theta_{Ref} = \tan^{-1}\left(-\frac{5}{7}\right)$$

$$= -.620$$



$$\boxed{2.52}$$

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4. $r = 6\cos\theta$ at $\underline{\theta = \frac{\pi}{6}}$ $\quad x = r\cos\theta \quad y = r\sin\theta$

$$\boxed{\frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}}$$

$$x = 6\cos^2\theta$$

$$\frac{dx}{d\theta} = -12\cos\theta\sin\theta$$

$$= -12\cos\left(\frac{\pi}{6}\right)\sin\left(\frac{\pi}{6}\right)$$

$$= -12\left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$$

$$= -3\sqrt{3}$$

$$y = \underline{6\cos\theta}\,\underline{\sin\theta}$$

$$\frac{dy}{dt} = 6\cos\theta\cos\theta + \sin\theta(-6\sin\theta)$$

$$= 6\cos^2\theta - 6\sin^2\theta$$

$$= 6\left(\cos\frac{\pi}{6}\right)^2 - 6\left(\sin\frac{\pi}{6}\right)^2$$

$$= 6\left(\frac{\sqrt{3}}{2}\right)^2 - 6\left(\frac{1}{2}\right)^2$$

$$= 6\left(\frac{3}{4}\right) - 6\left(\frac{1}{4}\right) = 3$$

$$\frac{dy}{dx} = \frac{3}{-3\sqrt{3}} = -\frac{1}{\sqrt{3}}$$

$$7 + 7\sin\theta = 21\sin\theta$$

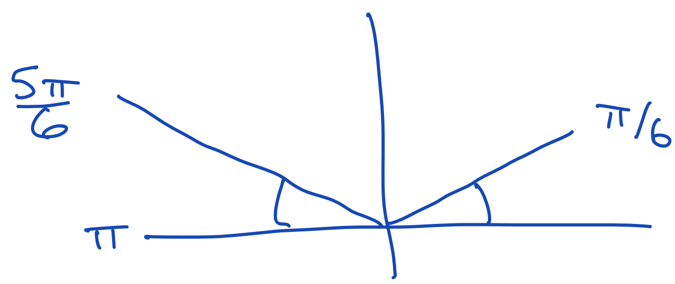
$$1 + \sin\theta = 3\sin\theta$$

$$-\sin\theta \qquad -\sin\theta$$

$$1 = 2\sin\theta$$

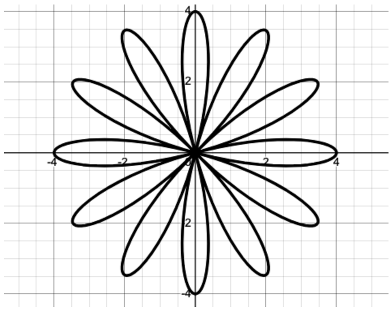
$$\frac{1}{2} = \sin\theta$$

$$\frac{\pi}{6} = \theta$$



Worksheet 10.4: Two polar area problems.

1. Find the area of one petal of the rose curve $r = 4\cos(6\theta)$



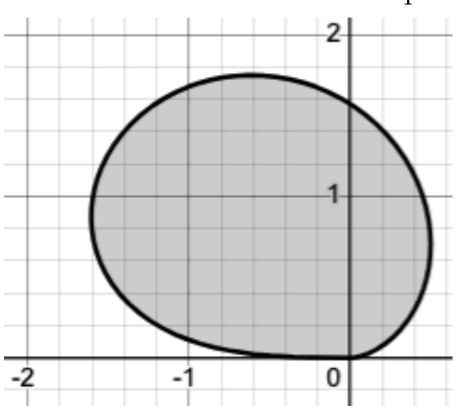
$$\frac{1}{2}\int_0^{\pi/6} [4\cos(6\theta)]^2\, d\theta$$

$$= \frac{1}{2}\int_0^{\pi/6} 16\underline{\cos^2(6\theta)}\, d\theta = 8\int_0^{\pi/6} \frac{1}{2}\left[1+\cos 12\theta\right] d\theta$$

$$4\left[\theta + \frac{1}{12}\sin 12\theta\right]_0^{\pi/6} = 4\left[\frac{\pi}{6} + \underbrace{\frac{1}{12}\sin 2\pi}_{0}\right] - 4\left[0-0\right]$$

$$= \frac{4\pi}{6} = \boxed{\frac{2\pi}{3}}$$

2. The blob below is the the graph of $r = \sqrt{\theta^2 \sin\theta}$ for $0 \le \theta \le \pi$. Find the area of the blob. Round to to 3 decimal places.



$$\frac{1}{2}\int_0^{\pi} \theta^2 \sin\theta\, d\theta$$

Tabular

$+\theta^2$	$\sin\theta$
-2θ	$-\cos\theta$
$+2$	$-\sin\theta$
0	$\cos\theta$

$$\frac{1}{2}\left[-\theta^2\cos\theta + 2\theta\sin\theta + 2\cos\theta\right]_0^{\pi}$$

$$\frac{1}{2}\left[-\pi^2\underbrace{\cos(\pi)}_{(-1)} + 2\pi\underbrace{\sin\pi}_{0} + 2\underbrace{\cos\pi}_{(-1)}\right] - \frac{1}{2}\left[0 + 0 + 2\underbrace{\cos 0}_{(1)}\right]$$

$$\frac{1}{2}\left[\pi^2 - 2\right] - 1 = 2.934\ldots$$

1. $\frac{2\pi}{3}$ 2. 2.934

11.1 Sequences

A **sequence** is an ordered list of numbers.

$$\{a_1, a_2, \dots, a_n, \dots\} \qquad \text{infinite in length.}$$

a_1 is the first **term**. We can denote the series by

$$\{a_n\} \qquad \text{or} \qquad \{a_n\}_{n=0}^{\infty}$$

0.0.1 Examples:

$\{2^n\}_{n=0}^{\infty} = \{1, 2, 4, 8, \dots\}$

$\underline{\{n^2\}}_{n=0}^{\infty} = \{0, 1, 4, 9, 16, \dots\}$

Note: A sequence can be thought of as a function with its domain restricted to the nonnegative integers.

An **alternating sequence** changes sign every term.

0.0.2 Examples:

$\{1, -1, 1, -1, \dots\}$

n	0	1	2	3	4	
$a_n = \dfrac{(-1)^n}{2^n}$	1,	$\frac{-1}{2}$,	$\frac{1}{4}$,	$\frac{-1}{8}$,	$\frac{1}{16}$,	$\dots$

$a_n = \dfrac{(-1)^{n+1}}{2^n}$ $\quad \dfrac{(-1)^{0+1}}{2^0}, \dfrac{(-1)^2}{2^1}, \dfrac{(-1)^3}{2^2}, \dfrac{(-1)^4}{2^3}, \dfrac{(-1)^5}{2^4}$

$-1, \frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, -\frac{1}{16}, \dots$

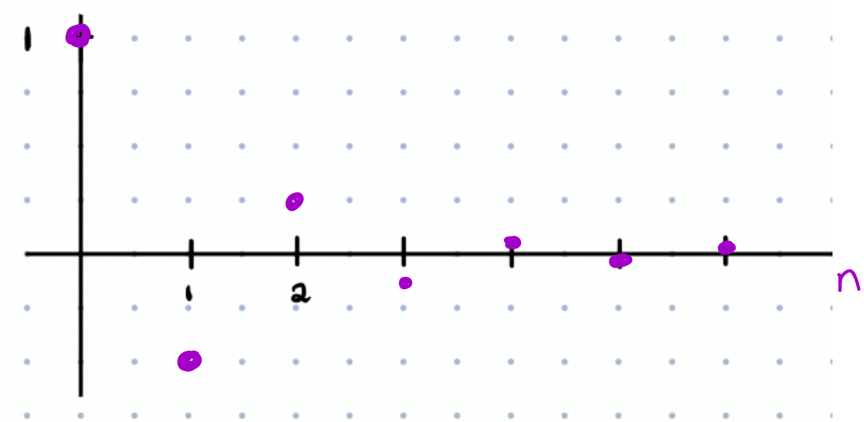
Note: In order to be considered an alternating sequence, the sign has to change every term. (e.g. $\{1, -2, -3, 4, -5, -6, 7, -8 - 9 \dots\}$ is not an alternating sequence.)

0.0.3 Example:

Graph the sequence $a_n = \dfrac{(-1)^n}{2^n}$

$a_0 = \dfrac{(-1)^0}{2^0} = \dfrac{1}{1} = 1 \quad a_1 = \dfrac{-1}{2}$

$a_2 = \dfrac{1}{4}$



Converges to 0.

To determine if a sequence $\{a_n\}$ converges or diverges, evaluate the limit

$$\lim_{n\to\infty} a_n$$

- if the limit exists, the sequence converges.
- if the limit does not exist, the sequence diverges.

0.0.4 Examples:

Examples of divergent series. Assume these start at $n = 1$

$a_n = \dfrac{n^2+1}{n}$ $\quad \{2, \frac{5}{2} = 2.5, \frac{10}{3} = 3.\overline{3}, \ldots\}$ diverges to ∞

$a_n = (-1)^n$ $\quad \{-1, 1, -1, 1, -1, 1, \ldots\}$ oscilleting. (not approaching a limit).

$a_n = \dfrac{3^n}{2^n + n}$ $\quad \displaystyle\lim_{n\to\infty} \frac{3^n}{2^n+n} = \infty$ faster.

Sequence Terminology

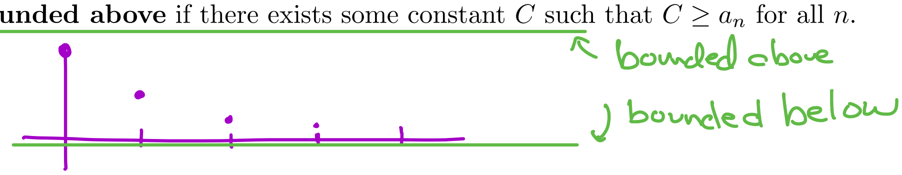
Let $\{a_n\}_{n=1}^{\infty}$ be a sequence. The sequence is ...

- **convergent** if $\lim_{n\to\infty} a_n$ exists.

- **divergent** if $\lim_{n\to\infty} a_n$ does not exist. This is usually because $a_n \to \infty$, $a_n \to -\infty$, or a_n oscillates or is erratic.

- **bounded above** if there exists some constant C such that $C \geq a_n$ for all n.

$a_n = \frac{1}{2n}$



- **bounded below** if there exists some constant C such that $C \leq a_n$ for all n.

- **bounded** if it is bounded above AND bounded below.

- **increasing** if $a_{n+1} > a_n$ for all $n \geq 1$

- **decreasing** if $a_{n+1} < a_n$ for all $n \geq 1$

- **monotonic** if it is either increasing or decreasing.

Monotonic Sequence Theorem: If a sequence is bounded and monotonic, then it converges.

Sequences Worksheet

In problems 1-17, an explicit formula for a is given. Determine if the sequence converges or diverges. If it converges, find $\boxed{\lim_{n\to\infty} a_n.}$

1. $a_n = \dfrac{n}{2n-1}$ c. to $\frac{1}{2}$

2. $a_n = \dfrac{3n+1}{n+2}$ $\to 3$

3. $a_n = \dfrac{4n^2+1}{n^2-2n+3}$

4. $a_n = \dfrac{3n^2+2}{n+4}$

5. $a_n = \dfrac{n+4}{2n^2+1}$

6. $a_n = \dfrac{\sqrt{n}}{n+1}$

7. $a_n = (-1)^n \dfrac{n}{n+1}$

8. $a_n = \dfrac{n\sin(n\pi/2)}{2n+1}$

9. $a_n = \dfrac{\sin(n\pi/2)}{n}$

10. $a_n = e^{-n}\cos n$

11. $a_n = \dfrac{e^n}{n^2}$ $e^n \leftarrow$ exponential (faster), $n^2 \leftarrow$ poly. $\lim_{n\to\infty} \dfrac{e^n}{n^2} = \infty$ diverges

12. $a_n = \dfrac{e^n}{2^n}$ $\lim_{n\to\infty} \dfrac{e^n}{2^n}$ $e^n \leftarrow$ faster, $2^n \leftarrow$ ∞

13. $a_n = \dfrac{(-\pi)^n}{4^n}$

14. $a_n = (\frac{1}{2})^n + 2^n$

15. $a_n = 1 + (0.9)^n$

16. $a_n = \dfrac{n^3}{e^n}$

17. $a_n = \dfrac{\ln n}{n}$

n^n

factorial $n!$

exponential c^n

polynomial or root n^c

logs $\log n$

c onstants

$n! = n(n-1)(n-2)\cdot 3\cdot 2\cdot 1$

n	$n!$
1	1
2	2
3	6
4	24
5	120
6	720
7	5040